Akoizo Original Practice - Form 2 - Answer key

Division B | Meteorology | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A station model pressure code of 984 means:

A. 984 mb

B. 1008.4 mb

C. 998.4 mb

D. 1098.4 mb

Correct: C

2. (2 points) Which air mass is warm and humid?

Unit: Ingredients for severe weather: air masses, fronts, jets, and terrain.
Worked-case data: On a spring afternoon in Oklahoma, a dryline lies near the Texas border, mT air with dew points near 20 °C is to its east, a strong jet streak passes overhead, and winds turn from southeast at the surface to west aloft. Assess the severe weather threat.
Separate investigation observations:
West of the line: Temperature 33 °C, dew point 2 °C.
East of the line: Temperature 31 °C, dew point 21 °C.
Wind shift: Southwest winds west of the line, southeast winds east.

A. Continental polar

B. Continental tropical

C. Continental arctic

D. Maritime tropical

Correct: D

3. (2 points) Damage spreading outward in a starburst pattern indicates:

Unit: Damaging winds and tornadoes.
Worked-case data: A damage survey finds trees in a field all blown down pointing outward from a central point, and a well-built house nearby lost its roof. Was this a tornado?
Separate investigation observations:
Damage path: A narrow path 300 m wide and 12 km long.
Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. A downburst

B. A tornado

C. Hail

D. A dryline

Correct: A

4. (2 points) Sleet forms when:

Unit: Precipitation, floods, winter storms, drought, and safety.
Worked-case data: A sounding shows below-freezing air from 3 km to the cloud top, a layer from 1.5 to 3 km at +3 °C, and a shallow layer from the ground to 300 m at -2 °C. What precipitation will reach the ground?
Separate investigation observations:
Rainfall: 150 mm fell in three hours over a small hilly basin.
Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. Hail falls from a supercell

B. Melted snow refreezes in a deep cold layer near the ground

C. Snow never melts

D. Rain freezes on contact

Correct: B

5. (2 points) In a METAR, “31/22” means:

A. Pressure 31.22 inches

B. Temperature 31 °C, dew point 22 °C

C. Wind 310° at 22 knots

D. Visibility 31 miles

Correct: B

6. (2 points) A dryline separates air masses that differ mainly in:

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A. Pressure only

B. Wind speed only

C. Moisture

D. Temperature

Correct: C

7. (2 points) The EF scale estimates wind speed from:

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Debris: Trees fell in different directions, converging toward the path center.
Homes: Well-built homes lost exterior walls; interior rooms remained.

A. Direct anemometer readings in the tornado

B. Radar reflectivity

C. Funnel width

D. Observed damage to indicators

Correct: D

8. (2 points) A tornado warning means:

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Separate investigation observations:
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Land surface: A wildfire burned the slopes last summer.
Stream gauge: Water rose 3 m in 90 minutes.

A. A tornado is occurring or imminent; take shelter now

B. Conditions may become favorable later

C. The storm has ended

D. Only hail is expected

Correct: A

9. (2 points) A long hodograph that curves clockwise near the ground indicates:

Unit: Reading weather data: station models, METARs, hodographs, and models.
Worked-case data: A station model shows 88 at upper left, 70 at lower left, a wind barb from the south with two long barbs and one short barb, and 096 at upper right. Decode it.
Separate investigation observations:
METAR: KDDC 151900Z 20030G45KT 7SM TSRA BKN030CB 29/21 A2972
Wind: From 200° at 30 knots gusting to 45 knots.
Weather: Thunderstorm with rain and cumulonimbus clouds.
Pressure: Altimeter 29.72 inHg and falling quickly.

A. Strong shear favoring supercells

B. No wind

C. Stable air

D. Calm conditions favoring fog

Correct: A

10. (2 points) Why do hurricanes weaken after landfall?

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A. Coriolis force becomes zero

B. They lose their warm-water energy source

C. Land air is too moist

D. The jet stream stops them

Correct: B

11. (2 points) A small round area of high reflectivity at a hook's tip with low CC suggests:

A. Ground clutter only

B. A gust front

C. A debris ball from a tornado

D. Light snow

Correct: C

12. (2 points) A "100-year flood" has what chance of occurring in any given year?

A. 100%

B. 10%

C. 0.1%

D. 1%

Correct: D

13. (4 points) Explain how to read wind direction from a wind barb.

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answer: The barb’s shaft points toward the direction the wind is coming from, and the feathers are on the end where the wind originates. (4 points).

14. (4 points) Explain how wind shear helps a thunderstorm last longer.

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answer: Shear tilts the updraft so falling rain and its cool downdraft occur away from the updraft, allowing warm inflow to continue. (4 points).

15. (4 points) Why can two tornadoes with similar winds get different EF ratings?

answer: Ratings depend on damage. A tornado crossing open fields damages few indicators and may be underrated compared with one striking well-built structures. (4 points).

16. (4 points) Explain why ice crystals grow at the expense of supercooled droplets.

answer: Saturation vapor pressure is lower over ice than over liquid water, so water vapor deposits onto ice while droplets evaporate. (4 points).

17. (4 points) Decode “TSRA” and “+RA” in a METAR.

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answer: TSRA is a thunderstorm with rain. +RA is heavy rain; the plus sign means heavy intensity. (4 points).

18. (4 points) Describe how a lake-effect snow band forms.

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answer: Cold air crosses a relatively warm lake, picks up heat and moisture over a long fetch, becomes unstable, and drops heavy snow downwind on shore. (4 points).